

Forecasting COP–USD

Oscar Díaz

EAFIT University

May 8, 2026

1 Data Studio 1:

- Results
 - Available Data Information
 - Main DFGCN Results
 - (DFGCN) Prediction for the Last 5 Years
 - Main Random Walk Results
 - (RW) Prediction for the Last 5 Years
- DM Test

2 Data Studio 2:

- Results
 - Available Data Information
 - Main DFGCN Results
 - (DFGCN) Prediction for the Last Year
- DM Test

Table: Descriptive statistics of exchange rates

Statistic	BRL/USD	CLP/USD	COP/USD
N	8360	8360	8360
μ	2.6786	601.0631	2397.5545
σ	1.5013	153.7802	998.2891
min	1.0045	368.7500	735.0000
$Q_{0.25}$	1.7335	482.3200	1802.5000
$Q_{0.50}$	2.2716	567.0000	2279.6450
$Q_{0.75}$	3.5974	690.8075	2989.4825
max	6.2931	1048.5000	5120.1500

Data Series



Figure: Time evolution of COP/USD and BRL/USD exchange rates (1993–2025).

Data Series



Figure: Time evolution of COP/USD and CLP/USD exchange rates (1993–2025).

(DFGCN) Full Prediction – 1 Week

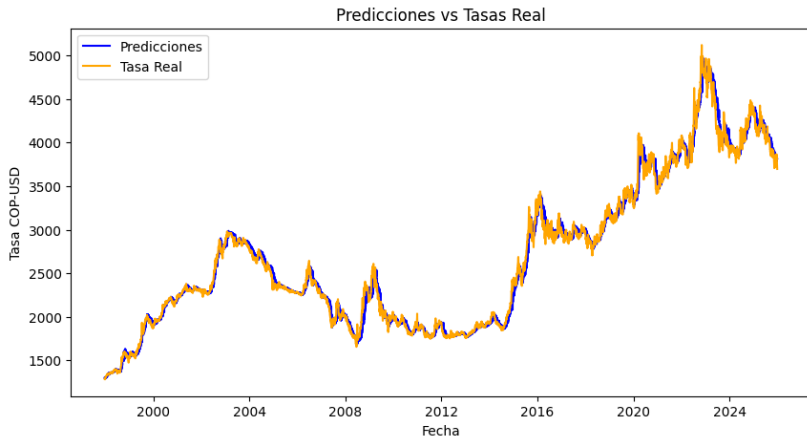


Figure: Time evolution, 1-week horizon, of predicted and actual COP/USD exchange rates (1997–2025).

(DFGCN) Full Prediction – 6 Months

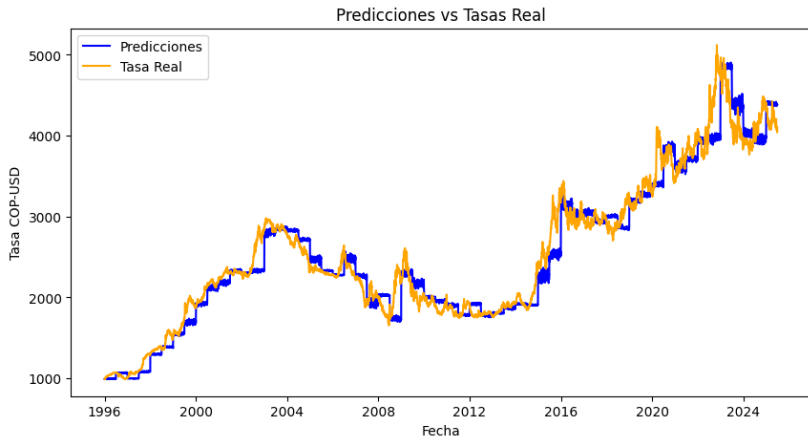


Figure: Time evolution, 6-month horizon, of predicted and actual COP/USD exchange rates (1996–2025).

(DFGCN) Full Prediction – 1 Year

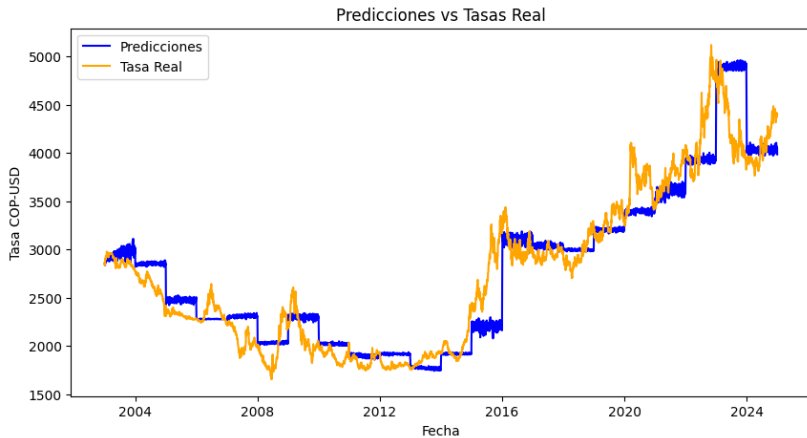


Figure: Time evolution, 1-year horizon, of predicted and actual COP/USD exchange rates (2002–2025).

(DFGCN) Last 5 Years Prediction – 1 Week

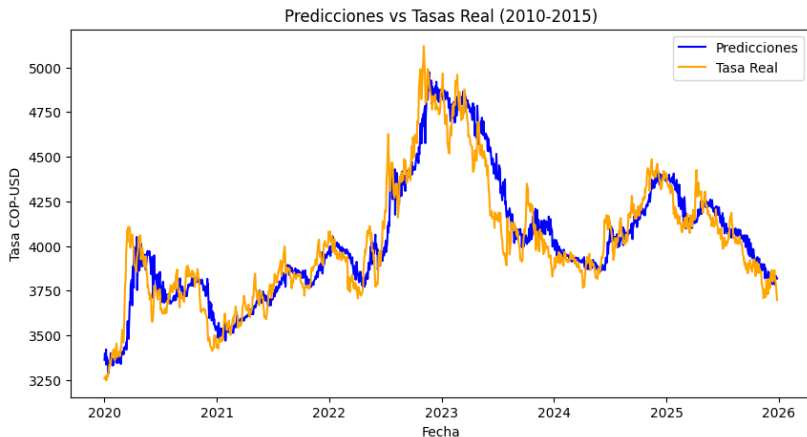


Figure: Time evolution (last 5 years), 1-week horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(DFGCN) Last 5 Years Prediction – 6 Months

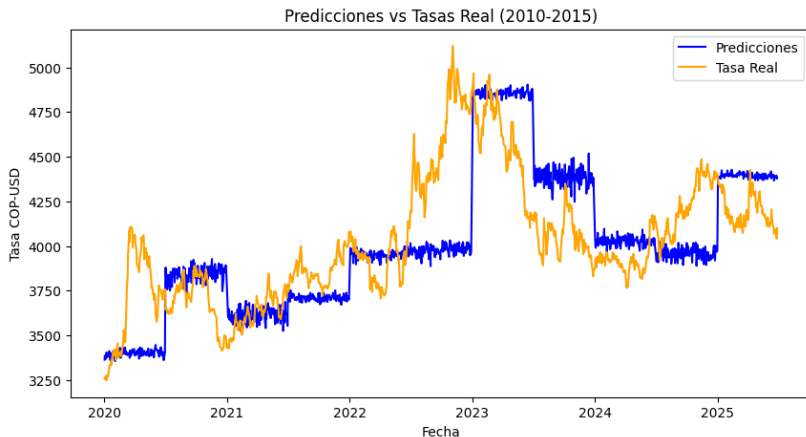


Figure: Time evolution (last 5 years), 6-month horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(DFGCN) Last 5 Years Prediction – 1 Year

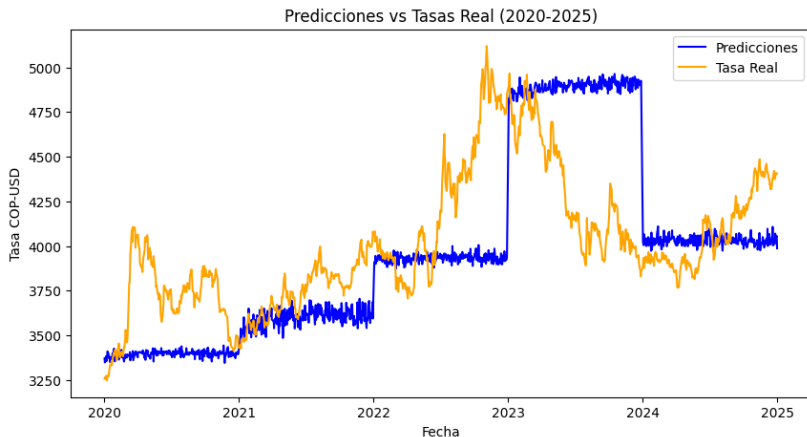


Figure: Time evolution (last 5 years), 1-year horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(RW) Full Prediction – 1 Week

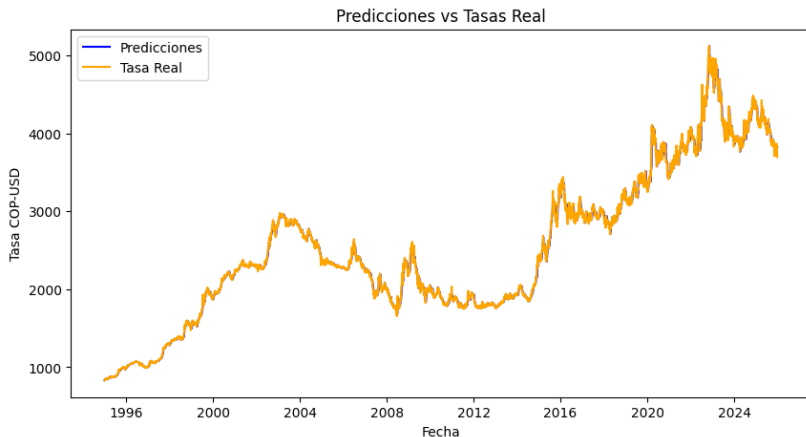


Figure: Time evolution, 1-week horizon, of predicted and actual COP/USD exchange rates (1993–2025).

(RW) Full Prediction – 6 Months

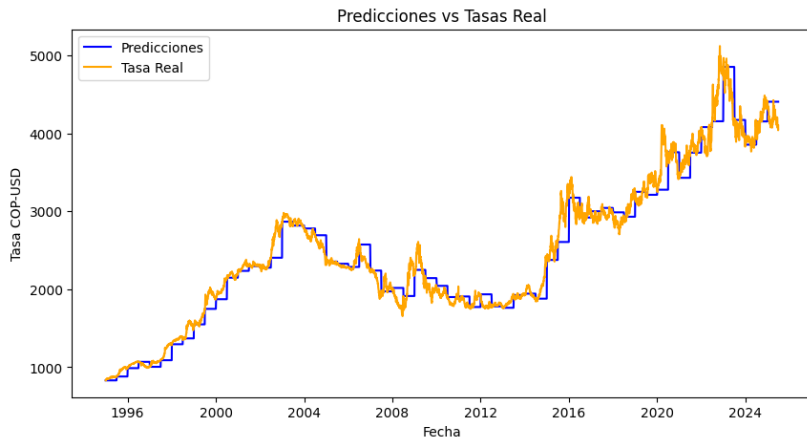


Figure: Time evolution, 6-month horizon, of predicted and actual COP/USD exchange rates (1994–2025).

(RW) Full Prediction – 1 Year

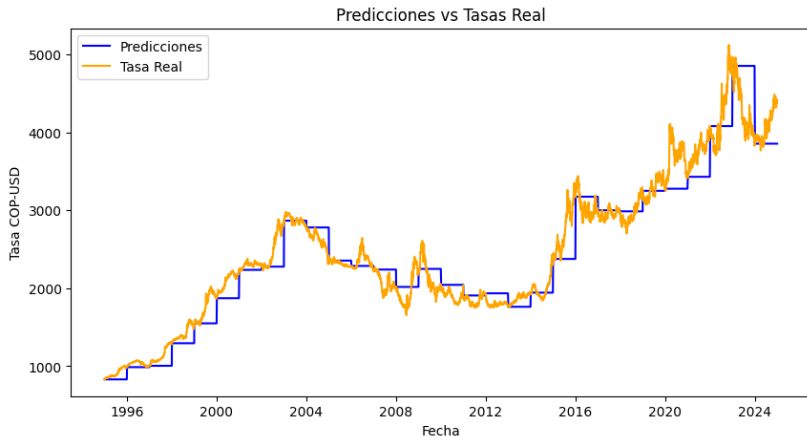


Figure: Time evolution, 1-year horizon, of predicted and actual COP/USD exchange rates (1995–2025).

(RW) Last 5 Years Prediction – 1 Week

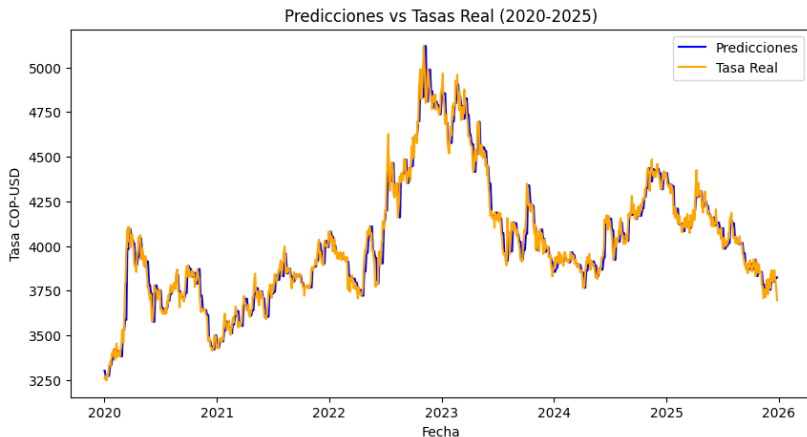


Figure: Time evolution (last 5 years), 1-week horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(RW) Last 5 Years Prediction – 6 Months

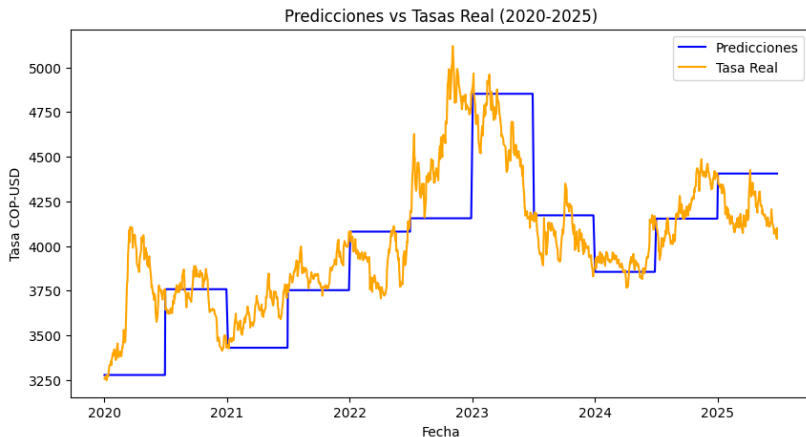


Figure: Time evolution (last 5 years), 6-month horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(RW) Last 5 Years Prediction – 1 Year

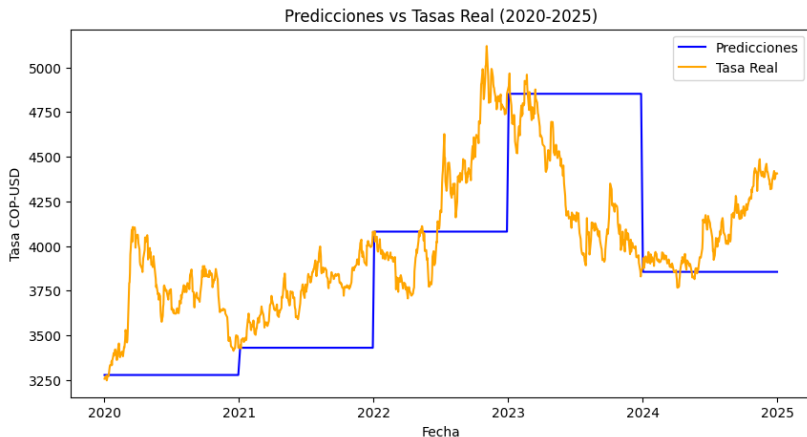


Figure: Time evolution (last 5 years), 1-year horizon, of predicted and actual COP/USD exchange rates (2020–2025).

Resultados DM DFGCN - Data ORates

Horizon	Train Size	DM	p-value	Accurate	Model	MAE DFGCN	MAE RW
1W	52 W	23.2346	0.0000	True	RW	55.1614	22.2344
1W	104 W	22.4450	0.0000	True	RW	55.5265	22.9130
1W	156 W	22.1492	0.0000	True	RW	55.9776	23.5772
1W	260 W	19.8500	0.0000	True	RW	55.1500	24.9938
2W	52 W	20.8619	0.0000	True	RW	61.1162	29.7533
2W	104 W	19.9429	0.0000	True	RW	61.3145	30.6602
1M	12 M	26.0330	0.0000	True	RW	73.5638	47.5440
1M	24 M	17.3665	0.0000	True	RW	74.7552	48.9794
2M	12 M	13.7725	0.0000	True	RW	89.4926	68.1135
2M	24 M	12.2734	0.0000	True	RW	89.4118	70.1412
3M	24 M	10.4935	0.0000	True	RW	101.7707	85.2823
3M	36 M	10.2420	0.0000	True	RW	104.1762	87.4573
6M	24 M	7.9778	0.0000	True	RW	135.7217	121.8195
6M	36 M	8.4656	0.0000	True	RW	141.6080	124.1599
1Y	3 Y	0.6114	0.5409	False	ND	178.0093	176.8441
1Y	5 Y	1.1546	0.2483	False	ND	186.1144	183.6278
1Y	10 Y	3.1167	0.0018	True	RW	198.0546	186.3763
2Y	6 Y	4.7261	0.0000	True	RW	293.7253	279.0585
2Y	10 Y	4.4875	0.0000	True	RW	308.4477	280.3887

Table: Diebold–Mariano test results for DFGCN (Data ORates)

Resultados DM MLR - Data ORates

Horizon	DM	p-value	Accurate	Model	MAE MLR	MAE RW
1Y	8.6254	0.0000	True	RW	208.5961	176.8441
1M	7.1117	0.0000	True	RW	61.5010	47.5440
1W	3.6228	0.0003	True	RW	26.0034	22.2344
2M	2.4386	0.0147	True	RW	116.5761	68.1135
2W	8.2046	0.0000	True	RW	36.4781	29.7533
3M	11.3713	0.0000	True	RW	107.5812	85.2823
6M	11.5685	0.0000	True	RW	173.8123	121.8195

Table: Diebold–Mariano test results for MLR (Data ORates)

Table: Descriptive statistics of market indicators

Statistic	Dif	VIX Index	BCOM Index	COP REGN Curncy
N	1827	1827	1827	1827
μ	4.0618	20.1430	95.5363	3918.1440
σ	1.9945	7.5880	16.1265	416.2342
min	-1.1440	11.5400	59.4795	3071.1800
$Q_{0.25}$	1.9255	15.1500	80.4577	3667.1900
$Q_{0.50}$	4.3970	18.1400	99.2939	3912.7000
$Q_{0.75}$	5.6520	22.9250	105.2976	4151.2700
max	7.5460	82.6900	136.6096	5120.1500

Data Series

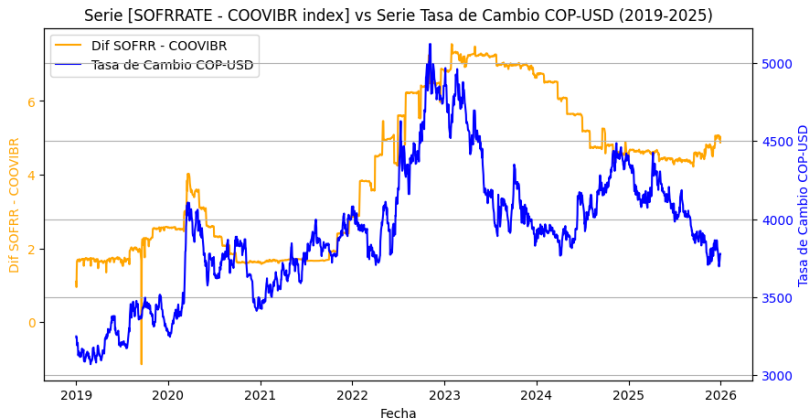


Figure: Time evolution of COP/USD and the difference of SOFR and COOVIBR rates (2019–2025).

Data Series

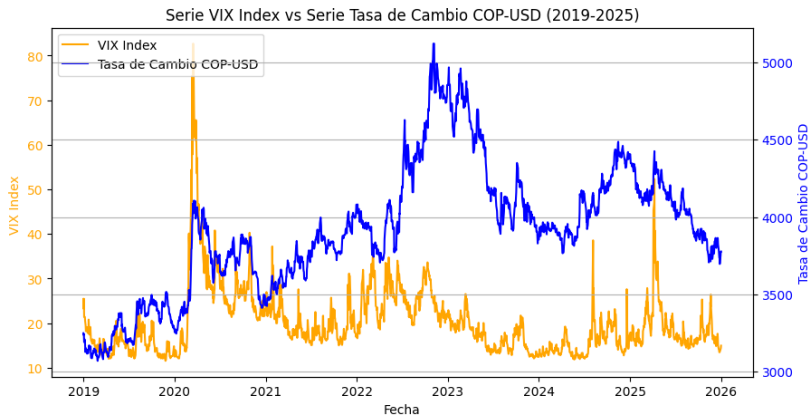


Figure: Time evolution of COP/USD and the VIX Index (2019–2025).

Data Series

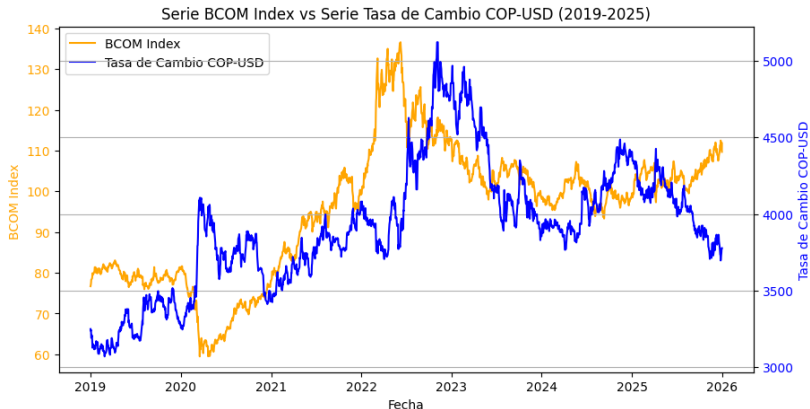


Figure: Time evolution of COP/USD and the BCOM Index (2019–2025).

(DFGCN) Full Prediction – 1 Week

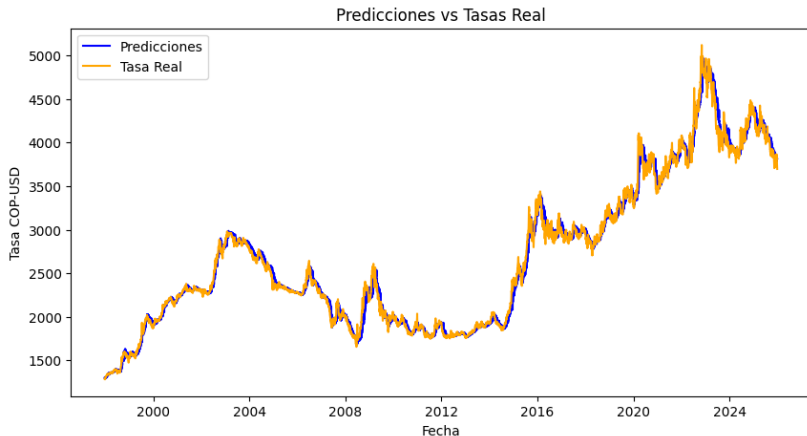


Figure: Time evolution, 1-week horizon, of predicted and actual COP/USD exchange rates (2019–2025).

(DFGCN) Full Prediction – 6 Month

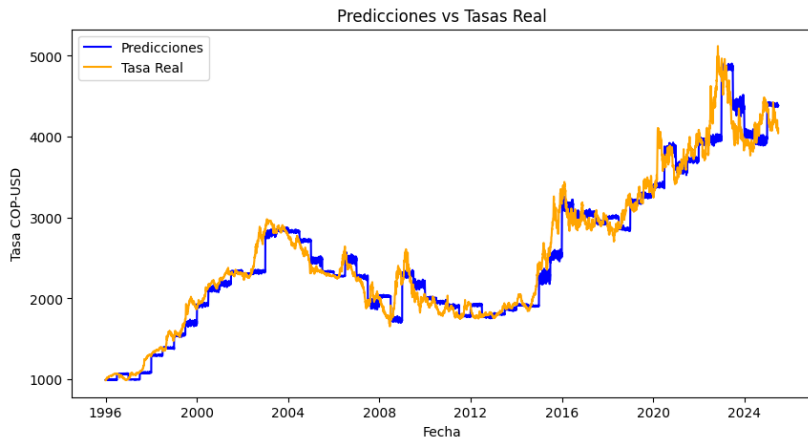


Figure: Time evolution, 6-month horizon, of predicted and actual COP/USD exchange rates (2019–2025).

(DFGCN) Full Prediction – 1 Year

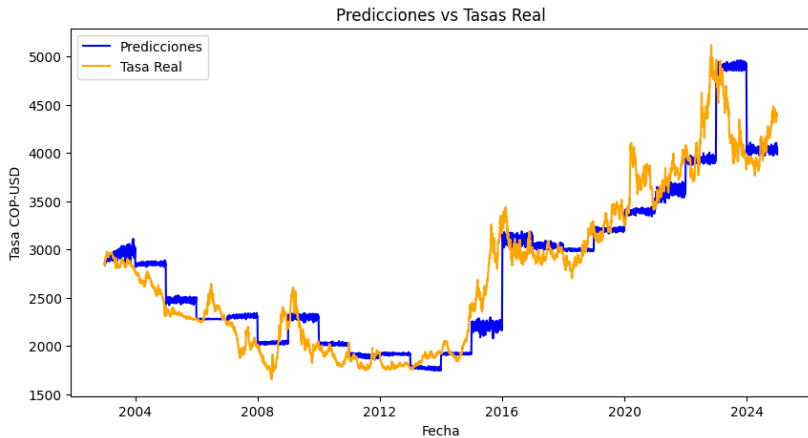


Figure: Time evolution, 1-year horizon, of predicted and actual COP/USD exchange rates (2019–2025).

(DFGCN) Last Year Prediction – 1 Week

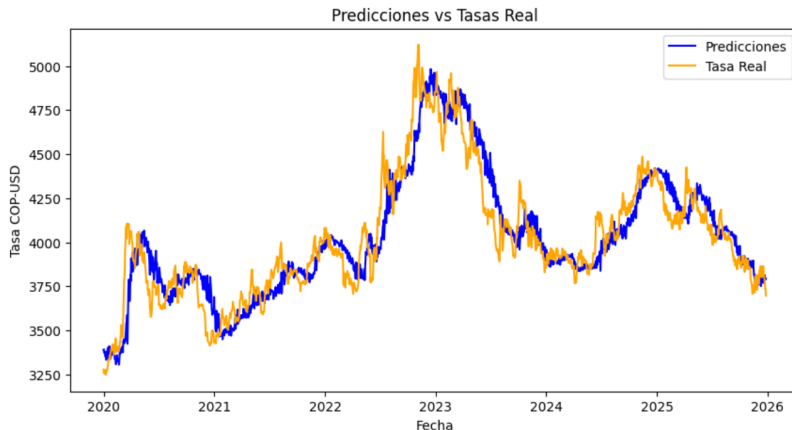
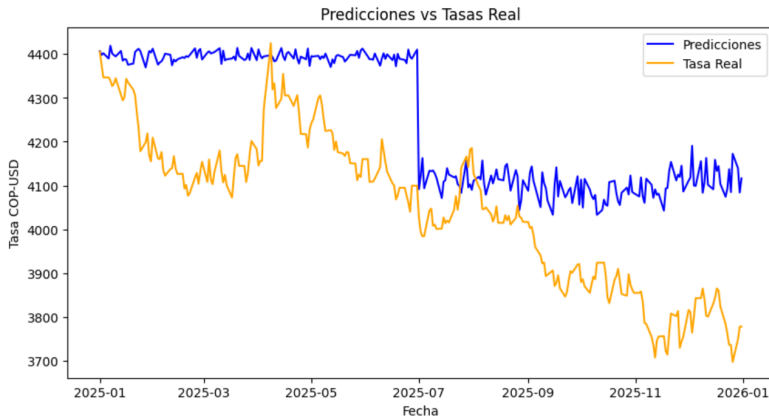


Figure: Time evolution (last year), 1-week horizon, of predicted and actual COP/USD exchange rates (2025).

(DFGCN) Last Year Prediction – 6 Months



-----Error-----

Mean error: 196.62852928190853

Std Desv error: 104.41426899640193

Median error: 216.35791015625

-----Absolute error-----

Mean abs error: 199.14768449922175

Std Desv abs error: 99.50619657749161

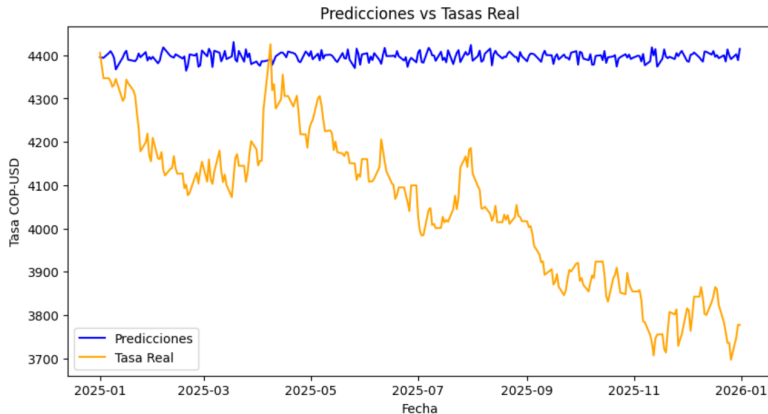
Median abs error: 216.35791015625

-----Estimation-----

Overestimation: 97.32%

Underestimation: 2.68%

(DFGCN) Last Year Prediction – 1 Year



-----Error-----

Mean error: 345.866619709351

Std Desv error: 175.27512855263177

Median error: 317.931640625

-----Absolute error-----

Mean abs error: 346.1398476786997

Std Desv abs error: 174.73284738060957

Median abs error: 317.931640625

-----Estimation-----

Overestimation: 99.23%

Underestimation: 0.77%

Resultados DM DFGCN - Data Profe

Horizon	DM	p-value	Accurate	Model	MAE DFGCN	MAE RW
1W	13.4042	0.0000	True	RW	103.9147	46.7774
1M	10.1108	0.0000	True	RW	134.9567	92.5711
2M	7.7869	0.0000	True	RW	163.4273	128.4538
3M	8.9455	0.0000	True	RW	175.0709	133.9126
6M	7.1539	0.0000	True	RW	247.7569	198.3234
1Y	0.2064	0.8365	False	ND	356.3232	368.6662

Table: Diebold–Mariano test results for DFGCN (Data Profe)

Resultados DM MLR - Data Profe

Horizon	DM	p-value	Accurate	Model	MAE MLR	MAE RW
1Y	7.0510	0.0000	True	RW	485.5266	368.6662
1M	7.1079	0.0000	True	RW	143.3316	92.5711
1W	1.1240	0.2611	False	ND	227.5177	46.7774
2M	7.8708	0.0000	True	RW	192.1601	128.4538
3M	6.8758	0.0000	True	RW	208.6834	142.0493
6M	10.0577	0.0000	True	RW	332.0317	198.3234

Table: Diebold–Mariano test results for MLR (Data Profe)